

Patients with a disease called chronic myeloproliferative neoplasm (MPN) produce abnormally high levels of certain blood cells or platelets in their bone marrow. In many affected patients, progenitor blood cells acquire a mutation in the calreticulin (*CALR*) gene that is passed on to descendent blood cells.

In wild-type (WT) individuals, random mutations can generate mutant *CALR* proteins in small numbers. However, researchers hypothesize that WT individuals can mount immune responses against mutant *CALR* proteins, preventing disease progression.

Researchers generated a peptide (CALRLong4) with the same amino acid sequence as a portion of the mutant *CALR* protein in patients with MPN. They also generated a random peptide with no similarity to mutant *CALR* (negative control). Lymphocytes and dendritic cells were isolated from WT individuals and exposed to one peptide. Enzyme-linked immune absorbent spot (ELISpot) was used to analyze these immune cells for cytokine IFN- γ secretion following peptide exposure (Figure 1).

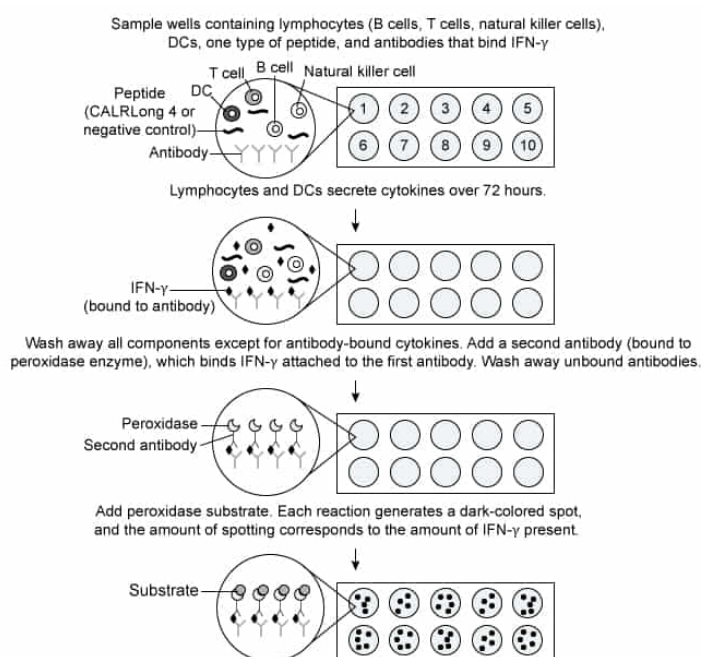


Figure 1 ELISpot procedure

Researchers want to use the ELISpot procedure (Figure 1) to test

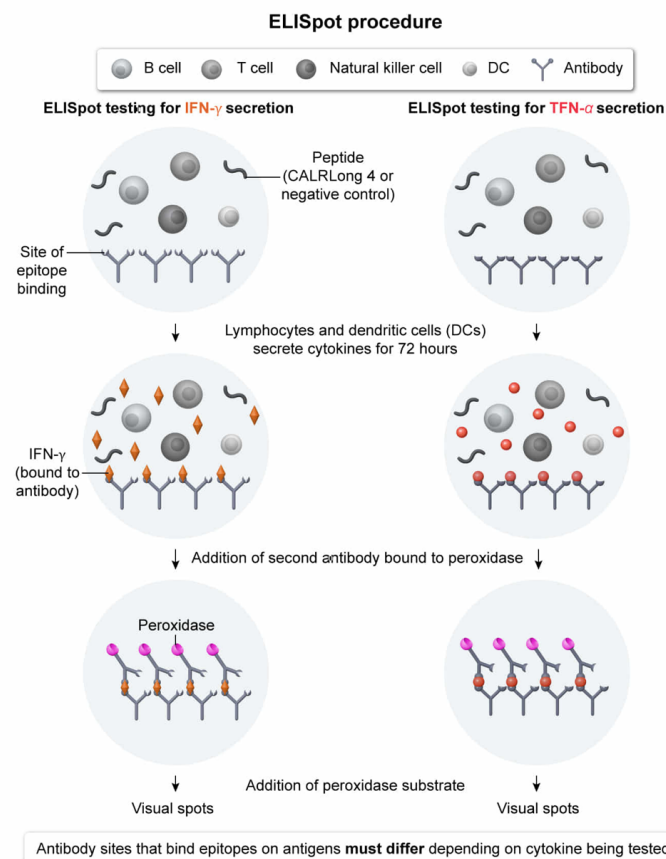
lymphocyte and dendritic cell samples from the WT individuals for secretion of another type of cytokine (TNF- α) upon exposure to these peptides. Given this requirement, which of the following factors from the original experimental procedure must be altered for this new experiment to function properly?

- ☐ A. Type of enzyme bound to second antibody [26%]
- ☐ B. Concentration of each antibody added to the wells [3%]
- ☒ C. Structure of antibody sites responsible for binding epitopes [66%]
- ☐ D. Length of time cells are incubated with the peptides and antibodies [3%]

Correct

66% answered correctly

Explanation:



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Antibodies are Y-shaped proteins that bind **antigens** (ie, foreign substances), allowing for better recognition and destruction of antigens

by **immune cells**. Antibodies contain **variable regions** that bind specific antigen regions called **epitopes**. In addition, antibodies bind **only one type** of antigen, and the antigen bound depends on the antibody's variable region structure. Accordingly, antibodies can be used to identify antigens in **molecular experiments**.

According to the passage, chronic myeloproliferative neoplasm (MPN) can occur when progenitor **blood cells** acquire a calreticulin (*CALR*) gene **mutation** passed to descendent blood cells. In WT individuals, random mutations may generate mutant CALR proteins in small numbers. However, researchers hypothesize that WT individuals mount immune responses against mutant CALR proteins, preventing disease progression.

The researchers generated two peptides: CALRLong4, which had the same structure as part of the mutant CALR protein in patients with MPN, and a negative control peptide. Lymphocytes and dendritic cells isolated from WT individuals were exposed to one of these peptides. ELISpot was then used to test these cells for secretion of **cytokine** IFN- γ (Figure 1).

As Figure 1 shows, ELISpot relies on **antibodies** that **recognize and bind a specific cytokine**. Initial antibodies bind the secreted cytokine, and then second antibodies bound to peroxidase **enzymes** bind the cytokines attached to the initial antibodies. By adding peroxidase substrate, the amount of cytokine present can be visualized (ie, via spots).

In this question, researchers want to use ELISpot to test WT lymphocytes and dendritic cells for secretion of another cytokine (TNF- α) upon peptide exposure. Because antibodies **bind only one type of antigen**, the **structure of the antibody sites** responsible for **binding epitopes** should be changed (ie, sites specific to TNF- α). The other options listed would *not* allow for TNF- α detection.

(Choice A) Using a different enzyme than peroxidase would *only* change the substrate used for visualizing the cytokine.

(Choice B) Changing the antibody concentrations would *only* alter the amount of *IFN- γ -specific* antibodies present.

(Choice D) Changing the incubation time for cells with peptides and antibodies would *only* alter how long cellular cytokine secretion and antibody binding occur before visualization via the enzyme.

Educational objective:

Antibodies are immune system proteins that have variable regions able to bind specific antigens, neutralizing these antigens and enhancing the responses of other immune cells.

Time spent:0 sec

Subject:

Biology

QID:403464

System:

Skin and Immune Systems

Patients with a disease called chronic myeloproliferative neoplasm (MPN) produce abnormally high levels of certain blood cells or platelets in their bone marrow. In many affected patients, progenitor blood cells acquire a mutation in the calreticulin (*CALR*) gene that is passed on to descendent blood cells.

In wild-type (WT) individuals, random mutations can generate mutant *CALR* proteins in small numbers. However, researchers hypothesize that WT individuals can mount immune responses against mutant *CALR* proteins, preventing disease progression.

Researchers generated a peptide (CALRLong4) with the same amino acid sequence as a portion of the mutant *CALR* protein in patients with MPN. They also generated a random peptide with no similarity to mutant *CALR* (negative control). Lymphocytes and dendritic cells were isolated from WT individuals and exposed to one peptide. Enzyme-linked immune absorbent spot (ELISpot) was used to analyze these immune cells for cytokine IFN- γ secretion following peptide exposure (Figure 1).

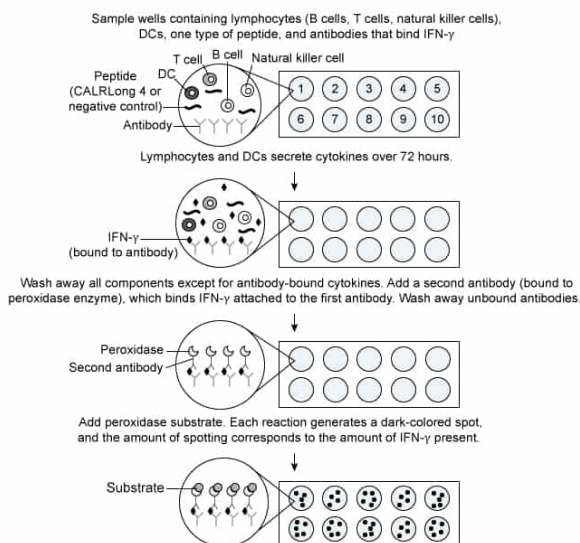


Figure 1 ELISpot procedure

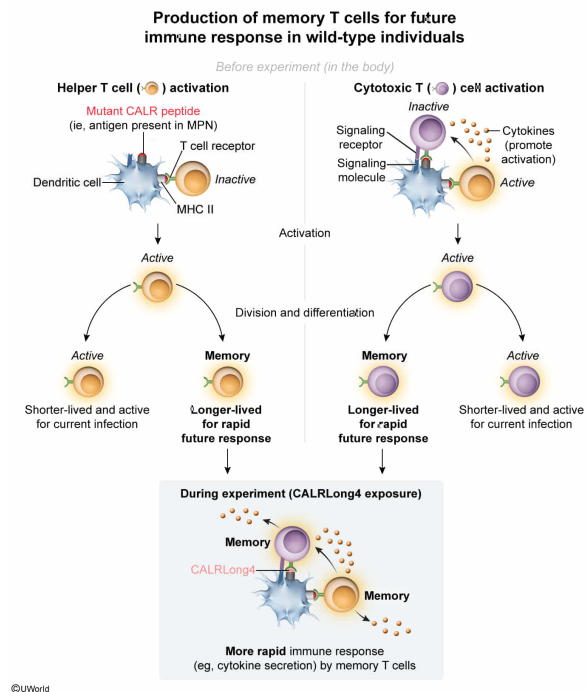
Many of the lymphocytes isolated from WT individuals that produced IFN- γ in response to CALRLong4 were found to be active memory T cells. Which of the following most likely explains this observation? Prior to the experiment, WT individuals had:

- ☐ A. many T cells that were eliminated via negative selection because they could not recognize *CALR* mutations causing MPN. [4%]
- ☐ B. some T cells that became activated by self-antigens produced from the WT *CALR* gene. [18%]
- ☐ C. many T cells that divided into antibody-secreting cells capable of recognizing mutant *CALR* proteins. [21%]
- ☒ D. some T cells that recognized and bound mutant *CALR* peptides similar or identical to those seen in patients with MPN. [56%]

Correct

57% answered correctly

Explanation:



The **adaptive immune system** consists of both **B** and **T cells**, **lymphocytes** that recognize *specific* foreign **antigens** and mount specialized immune responses against pathogens.

T cells bind foreign antigens displayed by **major histocompatibility complex (MHC) proteins** found on the surfaces of other cells (ie, other **immune cells**, infected cells). T cell types include:

- **Helper T cells**, which produce **cytokines** that function to activate other immune cells.
- **Cytotoxic T cells**, which induce **apoptosis** in cells infected by pathogens.

In addition, upon activation via antigen binding some T cells divide into **memory T cells** (eg, memory helper T cells, or memory cytotoxic T cells), which respond more rapidly to future **pathogen infection**.

In the passage, researchers exposed lymphocytes and dendritic cells from WT individuals to CALRLong4, a peptide with a structure identical to a portion of the mutant calreticulin (CALR) protein. This protein is found in many patients with MPN. The secretion of cytokine IFN- γ by immune cells in *response* to CALRLong4 was measured using ELISpot (Figure 1).

In the question, many WT lymphocytes that produced IFN- γ in response to CALRLong4 were active **memory T cells**, which are produced upon T cell *exposure* to *specific*

antigens. Therefore, prior to the experiment, WT individuals likely had some **T cells** that **recognized and bound mutant CALR peptides** similar or identical to those seen in **patients with MPN**, allowing for the formation of **memory** cells able to respond to mutant CALR peptides (ie, CALRLong4).

(Choices A and B) T cells initially produced in the bone marrow are transported to the thymus, a specialized **lymphoid** organ facilitating T cell maturation. Within the thymus, **negative selection** eliminates any T cells that recognize self-antigens (ie, molecules produced by healthy cells). Therefore, negative selection would likely *prevent* the presence of T cells capable of being activated by *self-antigens*, including those produced from the WT *CALR* gene. Furthermore, removing certain T cells via negative selection would *not* explain the presence of memory T cells, which are produced following *exposure* to a *specific* antigen.

(Choice C) B cells (*not* T cells) form antibody-secreting plasma cells upon antigen binding.

Educational objective:

T cells bind foreign antigens displayed by other cells (ie, immune cells, infected cells). Helper T cells release cytokines that enhance immune responses whereas cytotoxic T cells promote the apoptotic death of infected cells.

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Biology

QID:403465

System:
Skin and Immune Systems

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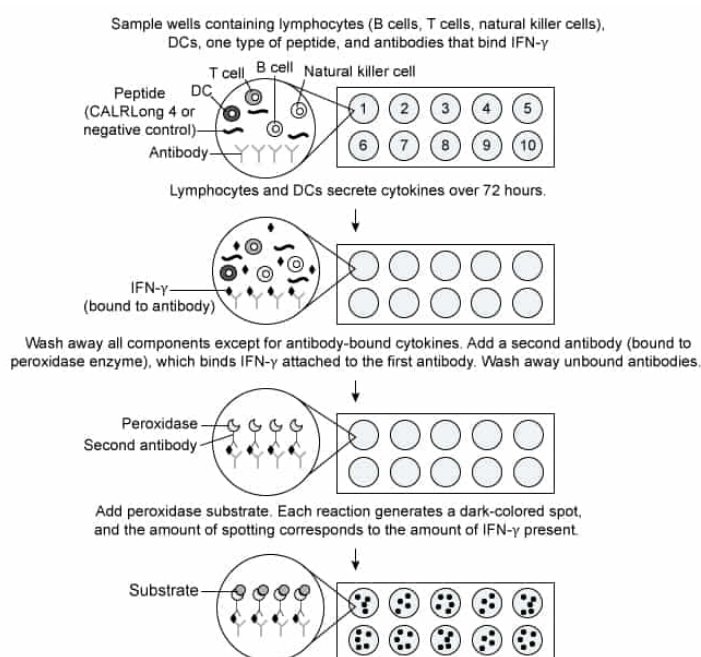
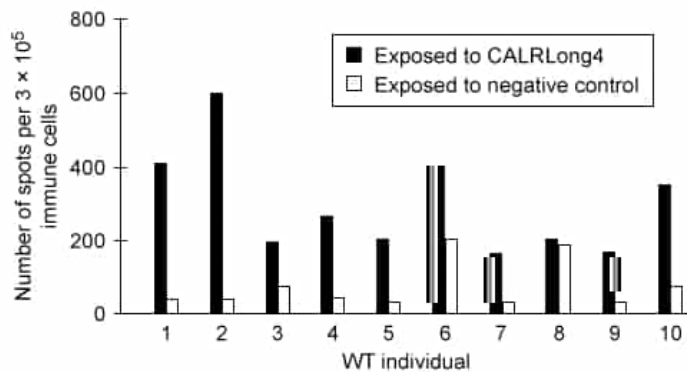


Figure 1 ELISpot procedure

Researchers quantified the ELISpot results to determine how

CALRLong4 and negative control peptides affected the amount of IFN- γ produced by isolated WT immune cells from different individuals. The results are shown below.



Which of the following statements is best supported by these results?

- ☐ A. The negative control peptide greatly inhibits IFN- γ production in all immune cells of WT individuals. [10%]
- ☒ B. Immune cells from different individuals have varying levels of IFN- γ secretion when exposed to either CALRLong4 or the negative control peptide. [73%]
- ☐ C. CALRLong4 elicits lower levels of IFN- γ secretion than the negative control peptide in WT immune cells. [2%]
- ☐ D. IFN- γ secretion differs significantly between immune cells from WT individuals and immune cells from individuals with high numbers of mutant CALR proteins. [13%]

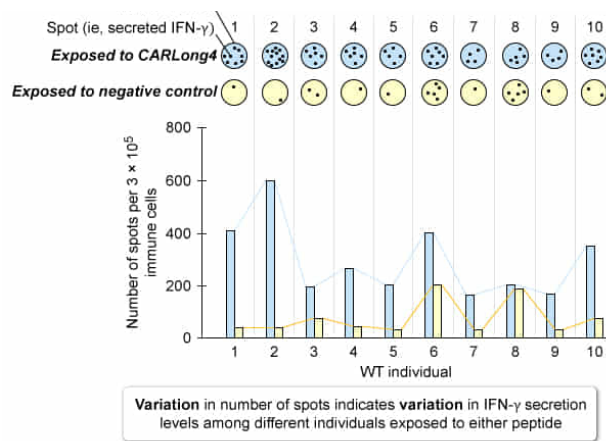
Correct

73% answered correctly

Explanation:

ELISpot analysis of IFN- γ secretion in WT immune cells exposed to different peptides





Antigens (ie, foreign molecules) can be recognized by **immune cells** (eg, lymphocytes, dendritic cells), eliciting an immune response that eliminates a pathogen or harmful protein. Upon interacting with an antigen, immune cells may secrete **cytokines** important to immune system function.

In the passage, researchers exposed lymphocytes and dendritic cells isolated from WT individuals to a mutant CALR peptide (**CALRLong4**) or a **negative control peptide**. According to the passage, WT individuals may express mutant CALR peptides in *small* numbers but are hypothesized to mount an immune response against these mutant peptides.

Researchers used ELISpot to test the WT immune cells for secretion of cytokine IFN-γ following peptide exposure (Figure 1). Specifically, each spot produced during ELISpot (ie, via an **enzymatic reaction** involving peroxidase-bound antibodies) represents the presence of IFN-γ, and the **amount of spotting corresponds to the amount of IFN-γ** that was secreted.

In this question, the ELISpot results for each individual were quantified. The question figure shows that the **number of spots** produced following immune cell exposure to either CALRLong4 or the negative control peptide **varied** among **different individuals**. Accordingly, this result indicates that **immune cells** from **different individuals** have **varying levels** of **IFN-γ secretion** when exposed to either peptide.

(Choices A and C) The question figure shows that IFN-γ secretion *still* occurred in WT immune cells exposed to the negative control, although

at *lower* levels than in WT immune cells exposed to CALRLong4. However, in certain cases IFN- γ secretion was *similar* between immune cells exposed to CALRLong4 and those exposed to the negative control. Therefore, the negative control likely did *not* greatly inhibit IFN- γ production in *all* immune cells of WT individuals.

(Choice D) The data on IFN- γ secretion levels in the question figure pertain to *only* WT individuals, who can contain *small* numbers of mutant CALR proteins. No data is presented in the question comparing the responses of these WT individuals to those of individuals with *high* numbers of mutant CALR proteins (eg, individuals with MPN).

Educational objective:

Antigens (ie, foreign molecules) can be recognized by immune cells, which can then mount an immune response against the source of these antigens (eg, pathogen).

Time spent:0 sec

Subject:
Biology

QID:403466

System:
Skin and Immune Systems

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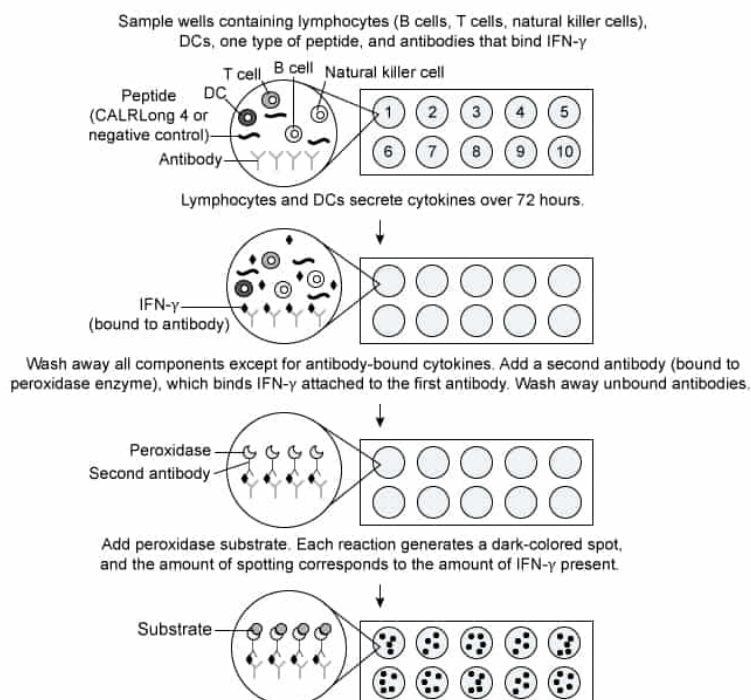


Figure 1 ELISpot procedure

Which of the following describes an appropriate positive control to include in the passage experiment?

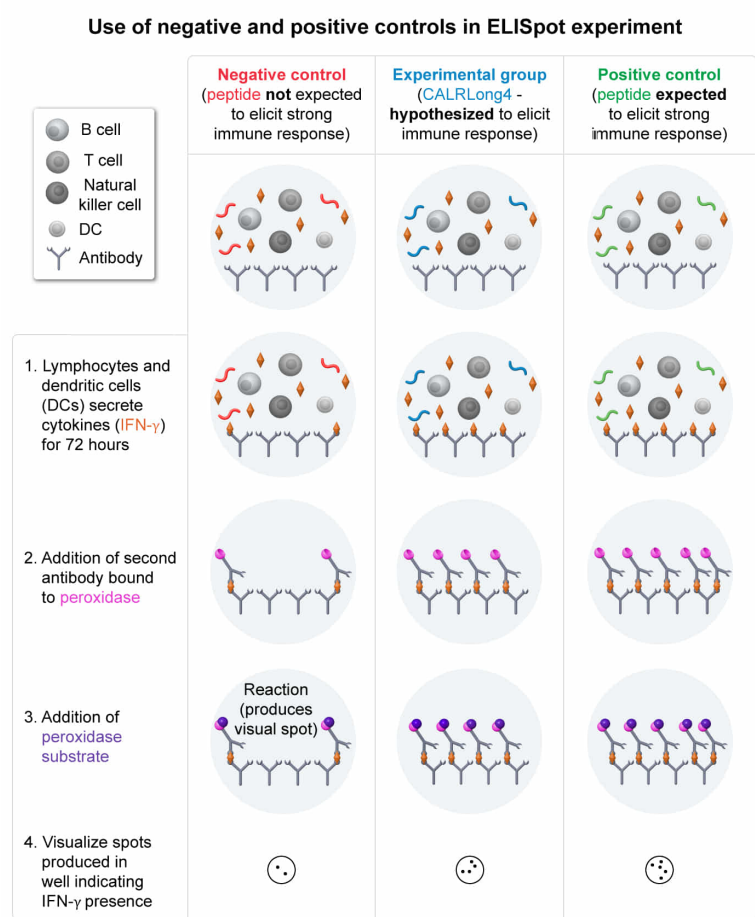
- ✔

A. A peptide previously shown to elicit a strong immune response [56%]
- B. A sample of immune cells from a patient with a disease unrelated to MPN [4%]
- C. A second randomly generated peptide that differs from the negative control [3%]
- D. A sample of immune cells from a patient exhibiting symptoms of MPN [35%]

Correct

56% answered correctly

Explanation:



In an experiment, a **positive control** is the experimental factor known to produce an **expected result** (ie, known effect on the **dependent variable**). By using positive controls, researchers can determine if the experiment is **functioning properly**. For example, if the positive control does *not* produce the expected result, then an *issue* with the experimental setup would be indicated (eg, nonfunctional reagents).

In addition, the positive control results can be compared to those of the experimental group(s), allowing researchers to better determine the level to which the independent variable impacted the dependent variable (ie, level of similarity between experimental group results and positive control results).

In the passage, researchers isolated lymphocytes and dendritic cells from WT individuals and exposed them to either a mutant calreticulin (CALR) peptide (CALRLong4) or a randomly generated **negative control** peptide. Next, they tested the immune cells for **cytokine** IFN- γ secretion using ELISpot, as shown in Figure 1.

If researchers were to add a **positive control** to this experiment, the most appropriate choice would be a **peptide** previously shown

to **elicit a strong immune response** (eg, **high** cytokine secretion levels from immune cells). This positive control would be expected to cause *high* IFN- γ secretion levels (dependent variable) if the experiment is working properly. Therefore, the effect of CALRLong4 (independent variable) on IFN- γ secretion could then be compared to the effect of the positive control to better assess the level of immune cell response to this mutant CALR peptide.

(Choices B and D) Including immune cells from a patient with a disease unrelated to MPN or a patient exhibiting MPN symptoms would *only* allow researchers to examine IFN- γ secretion in additional types of individuals upon peptide exposure.

(Choice C) The negative control is an experimental factor that is *not* expected to produce an effect on the dependent variable and can help *ensure* that **confounding factors** (ie, variables *other than* those being tested) are *not* impacting the results. Another randomly generated peptide could serve as an *additional* negative control because it would *not* be expected to yield a strong immune response.

Educational objective:

Positive controls, or experimental factors known to produce an expected experimental result, allow researchers to validate that an experimental setup is functioning properly (ie, that changes in the dependent variable can be properly detected).

Time spent:0 sec

Subject:

Biology

QID:403467

System:

Skin and Immune Systems